
CLEAN FEATURE ANCHORING: MITIGATING THE ACCURACY-ROBUSTNESS TRADE-OFF BY SELF-DISTILLING A NATURALLY TRAINED MODEL

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ABSTRACT

Adversarial training significantly improves adversarial robustness, but robust models attain substantially lower standard accuracy than naturally trained ones. This accuracy gap has motivated a line of work that brings a naturally trained network into adversarial training. Existing methods distill its logits into the robust model or add its predictions as an auxiliary regularizer beside the label loss. We also employ a naturally trained network, but we exploit it far more directly: the label loss is discarded entirely and the backbone is trained against the natural network alone. In this paper, we propose Clean Feature Anchoring (CFA), which anchors the adversarial feature of the student to the clean feature of the teacher rather than to its logits. We prove that this single objective bounds both fidelity to the teacher and local stability of the student up to a constant factor, so that no balancing coefficient is required. We further assign the attack radius per sample according to the input sensitivity of the loss under a fixed total budget, approximately equalizing the first-order loss change across examples. Experimental results show that CFA recovers standard accuracy without sacrificing robustness. Particularly, CFA attains 62.17% standard accuracy at 28.86% AutoAttack accuracy on CIFAR-100 with the ResNet-18 architecture, and consistent improvements are observed on CIFAR-10 and Tiny-ImageNet.

1 INTRODUCTION

Adversarial training remains the most reliable defense against norm-bounded attacks (Madry et al., 2017; Athalye et al., 2018; Croce & Hein, 2020), but its robustness comes with lower accuracy on clean inputs. This trade-off is partly intrinsic to the training objective (Tsipras et al., 2018; Zhang et al., 2019) and becomes especially restrictive for compact networks. Existing approaches improve the target seen during adversarial training: some replace one-hot labels with softened or self-distilled targets (Pang et al., 2020; Tack et al., 2022; Wu et al., 2024), while adversarial distillation transfers predictions from a robust teacher (Goldblum et al., 2020; Zhu et al., 2021; Zi et al., 2021; Huang et al., 2023; Lee et al., 2025). The latter can be effective, but presupposes a model that has already paid the cost of robust training.

A naturally trained copy of the student has the complementary property: it is inexpensive and holds the clean accuracy that adversarial training loses, but it is itself non-robust. Prior methods use such a model as auxiliary guidance. ARREST preserves its representations alongside a label objective (Suzuki et al., 2023); DP-FAT rescales its logits per example (Cho & Kim, 2026); and B-MTARD balances it against a second, robust teacher (Zhao et al., 2024). These designs raise a more basic question: *can a natural teacher provide the sole supervision for adversarial training, without transferring its fragility?*

Our approach separates the teacher’s role as a clean target from the student’s response to perturbations. The teacher supplies a representation at the clean input; the student learns to preserve it throughout the perturbation set, without matching the teacher at perturbed inputs. We instantiate this principle as **Clean Feature Anchoring** (CFA): adversarial training uses feature regression as

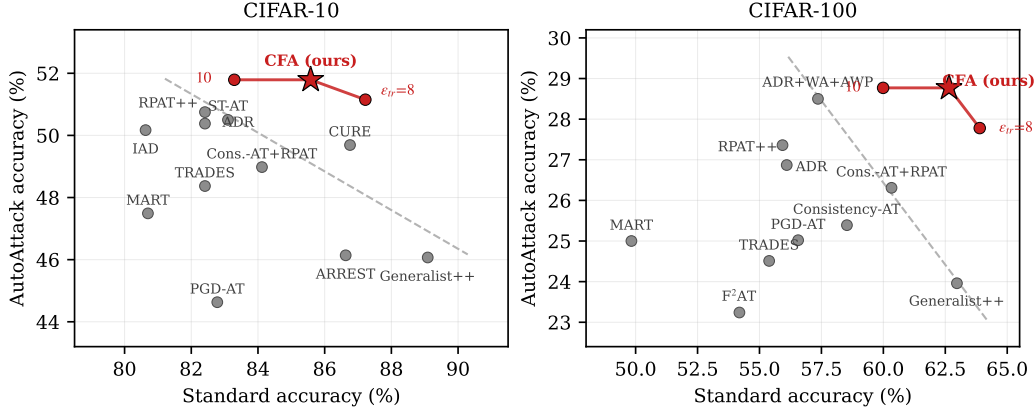


Figure 1: Clean–AutoAttack trade-off on CIFAR-10 and CIFAR-100 with ResNet-18 at $\epsilon = 8/255$. Grey points are published results (Table 26); the CFA curve varies the training radius, and the star marks the configuration used elsewhere.

the entire backbone objective and retains the teacher’s classifier. Softening probability targets improves robustness in our experiments, but direct regression offers a competitive alternative without temperature selection. Logit regression is also effective; the central finding is that clean-target regression can replace, rather than supplement, label supervision.

This single target has a useful structural property. Let L denote the worst-case distance from the student feature to the clean teacher feature, F the clean teacher–student discrepancy, and O the diameter of the student’s features over the perturbation ball. We show that $L \leq F + O \leq 3L$. Thus the same objective controls teacher fidelity and local variation without balancing separate penalties. The result does not certify robust classification, but it explains why removing the label loss does not leave stability unconstrained.

We further address a mismatch left by a uniform attack radius. The radius is measured in pixels, whereas CFA measures displacement in feature space, so equal pixel budgets can induce very different loss changes across examples. Our sensitivity-matched radius assigns less budget to locally sensitive examples and more to insensitive ones while preserving the batch total. This allocation improves both axes over the uniform-radius anchor and introduces no additional loss term.

CFA achieves competitive clean–robust trade-offs across CIFAR-10, CIFAR-100, and Tiny-ImageNet. On CIFAR-100 with ResNet-18 it reaches 62.17% clean and 28.86% AutoAttack accuracy (Figure 1). Controlled experiments show that the gain persists without weight averaging or adversarial weight perturbation, and that substituting label cross-entropy under matched training conditions reduces both clean and robust accuracy. Varying the natural-teacher checkpoint reveals that increasing teacher clean accuracy can reduce student clean accuracy while improving robustness. Changes in the teacher’s class separation are also reflected in the student, suggesting that teacher selection should consider the resulting student trade-off.

Our contributions are threefold:

- We show empirically that a natural teacher can supply the sole target for adversarial training. CFA realizes this with clean feature regression and an inherited classifier, without a label loss or temperature selection during robust training.
- We characterize how a fixed anchor controls fidelity and local feature variation, and develop a sensitivity-based radius allocation whose benefit persists when the radius distribution is controlled.
- Across three datasets and two architectures, we evaluate transfer beyond label training and recipe effects, and show that higher teacher clean accuracy need not improve student clean accuracy.

2 RELATED WORK

Adversarial training and its targets. PGD adversarial training optimizes a model on worst-case inputs (Madry et al., 2017); TRADES separates clean prediction from local consistency (Zhang et al., 2019), and MART reweights misclassified examples (Wang et al., 2020). Subsequent work improves generalization through label smoothing, consistency regularization, weight averaging, and adversarial weight perturbation (Pang et al., 2020; Tack et al., 2022; Rice et al., 2020; Wu et al., 2020). ADR replaces one-hot labels with targets produced by an annealed momentum teacher (Wu et al., 2024). These approaches motivate our study of how the supervisory target affects the clean-robust trade-off.

Adversarial distillation. ARD transfers a robust teacher’s predictions during adversarial training (Goldblum et al., 2020). RSLAD emphasizes robust soft labels (Zi et al., 2021), IAD discounts unreliable teacher guidance (Zhu et al., 2021), and AdaAD uses the teacher in both adversarial-example generation and student optimization (Huang et al., 2023). IGDM adds the teacher’s local input geometry to logit-based distillation (Lee et al., 2025). These methods are designed to inherit robustness from an adversarially trained teacher. In contrast, our teacher is trained only on clean data, and its clean representation remains fixed during the student’s inner maximization.

Guidance from naturally trained models. LBGAT jointly trains a natural branch and aligns adversarial logits to it (Cui et al., 2021). ARREST starts from a naturally pretrained model and regularizes latent representations while retaining adversarial cross-entropy and noisy replay (Suzuki et al., 2023). HAT uses a natural helper to label examples outside the training ball (Rade & Moosavi-Dezfooli, 2022), DP-FAT distills direction-preserving logits (Cho & Kim, 2026), and B-MTARD combines clean and robust teachers with adaptive temperature and loss balancing (Zhao et al., 2024). CFA shares the use of clean natural guidance, but differs in making a fixed clean representation the sole target throughout the adversarial-training stage.

Sample-dependent perturbation budgets. IAAT and CAT adapt the training radius to sample difficulty, while MMA directly increases an input-space margin (Balaji et al., 2019; Cheng et al., 2022; Ding et al., 2020). Our allocation has a different source: it redistributes a fixed batch budget according to the input sensitivity of the loss being optimized. The comparison is therefore not between adaptive and uniform radii alone, but between the signals used to assign the same budget.

3 ANALYSIS

A natural teacher offers high clean accuracy despite being vulnerable to adversarial perturbations. We examine what fixed signal it can provide for the student’s adversarial training. We compare probability-space supervision with direct regression, characterize the resulting anchor objective, and then vary the teacher checkpoint to examine what the student inherits.

3.1 PRELIMINARIES

Let $x \in \mathcal{X} = [0, 1]^D$ and $B(x, \epsilon) = \{x' \in \mathcal{X} : \|x' - x\|_\infty \leq \epsilon\}$. We decompose a network as $f = h \circ \Phi$, where $\Phi : \mathcal{X} \rightarrow \mathbb{R}^d$ is a continuous feature map and $h : \mathbb{R}^d \rightarrow \mathbb{R}^C$, $h(z) = Wz + b$, is the classifier for C classes. We use subscripts t and s for the frozen natural teacher and the student. Both networks share the same architecture, with fixed network statistics for the analysis; feature distances are Euclidean unless stated otherwise.

3.2 DO NATURAL TEACHERS NEED SOFTENED TARGETS?

Can the natural teacher’s concentrated clean predictions provide effective adversarial supervision if softened? Motivated by soft-label robust distillation (Goldblum et al., 2020; Zi et al., 2021; Wu et al., 2024; Zhao et al., 2024), we first vary target concentration before changing the representation being transferred. Let

$$q_t^\tau(x) = \text{softmax}(f_t(x)/\tau), \quad p_s^\tau(x') = \text{softmax}(f_s(x')/\tau). \quad (1)$$

We compare softening only the teacher target with conventional shared-temperature KD:

$$D_{\text{KL}}(q_t^\tau(x) \| p_s^1(x')), \quad \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')). \quad (2)$$

As $\tau \rightarrow \infty$, the teacher target approaches a uniform distribution in both cases, but the scaled shared-temperature loss retains logit information. For fixed logits and $\bar{f} = f - C^{-1} \mathbf{1} \mathbf{1}^\top f$,

$$\lim_{\tau \rightarrow \infty} \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')) = \frac{1}{2C} \|\bar{f}_t(x) - \bar{f}_s(x')\|_2^2. \quad (3)$$

This limit motivates direct regression onto clean teacher logits; we also examine regression onto the full clean feature, which defines

$$\mathcal{L}_{\text{anchor}}(x) = \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2. \quad (4)$$

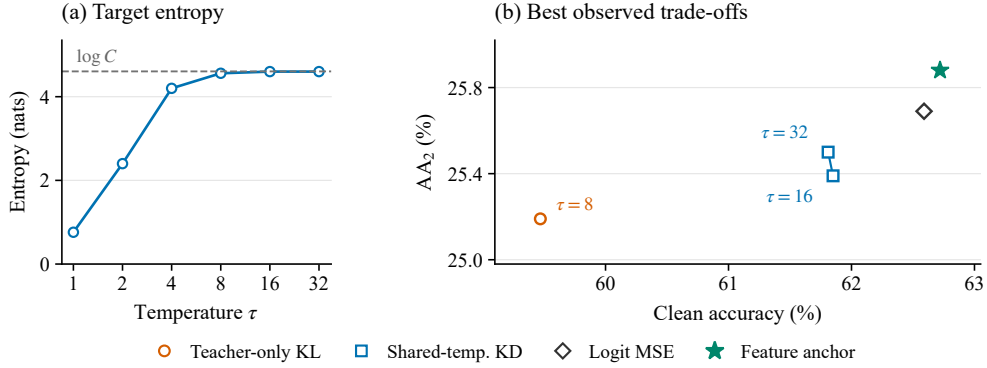


Figure 2: Target comparison on CIFAR-100 (ResNet-18). (a) Teacher-target entropy approaches the uniform value as τ increases. (b) Nondominated settings within each KD variant, alongside fixed-head logit and feature regression. AA₂ combines APGD-CE and APGD-T. The full temperature sweep is reported in Table 9.

We train for 50 epochs at $\epsilon = 8/255$, without weight averaging or AWP. Across the full sweep, teacher-only softening improves robustness until the target becomes nearly uniform, but changes clean accuracy little. Shared-temperature KD avoids this turnover and approaches logit regression (Figure 2). Feature regression is only 0.13 clean and 0.19 AA₂ points higher: both regression targets are effective without temperature selection, but this does not establish feature superiority. Moreover, KD updates the classifier whereas regression uses a fixed head, so this comparison evaluates training designs rather than isolating the target alone.

These results support direct regression to clean teacher targets in either space. We use features to preserve the representation supplied to the inherited classifier, including directions that logit regression need not constrain, and next examine what this single target controls.

3.3 WHAT THE ANCHOR CONTROLS

For a fixed x and $B = B(x, \epsilon)$, define the worst-case anchor distance, clean fidelity, and student variation as

$$L = \max_{x' \in B} \|\Phi_s(x') - \Phi_t(x)\|_2, \quad F = \|\Phi_s(x) - \Phi_t(x)\|_2, \quad O = \max_{x', x'' \in B} \|\Phi_s(x') - \Phi_s(x'')\|_2. \quad (5)$$

Proposition 1 (Fixed-anchor decomposition). *For every $x \in \mathcal{X}$ and $\epsilon \geq 0$,*

$$L \leq F + O \leq 3L. \quad (6)$$

Proof. For any $x' \in B$, the triangle inequality gives $\|\Phi_s(x') - \Phi_t(x)\|_2 \leq \|\Phi_s(x') - \Phi_s(x)\|_2 + F \leq O + F$; maximizing over x' yields $L \leq F + O$. Moreover, $x \in B$ implies $F \leq L$, and, for any $x', x'' \in B$, applying the triangle inequality through $\Phi_t(x)$ gives $\|\Phi_s(x') - \Phi_s(x'')\|_2 \leq 2L$. Hence $O \leq 2L$ and $F + O \leq 3L$. $\square$

If $L \leq \delta$, then the clean student feature is within δ of the teacher and the student’s feature diameter over the ball is at most 2δ , without assuming that the teacher itself is stable. The inherited linear head also bounds logit deviation by $\|W_t\|_{2 \rightarrow 2}L$, although classification depends on the teacher’s clean margin. The same argument applies to norm-based logit regression: it explains joint control by a fixed target, not a feature-specific advantage. Finite-step PGD does not certify L .

3.4 WHAT DOES THE NATURAL TEACHER TRANSFER?

We vary only the natural-teacher checkpoint along one training trajectory, using it for student initialization, feature targets, and classifier inheritance. We ask whether higher teacher clean accuracy translates into higher student clean accuracy, and whether changes in class geometry are preserved. We measure S_w/S_b on unit-normalized clean features: S_w is the mean squared distance to the corresponding class mean, and S_b is the mean squared distance of class means to the global mean. A smaller ratio indicates stronger class separation.

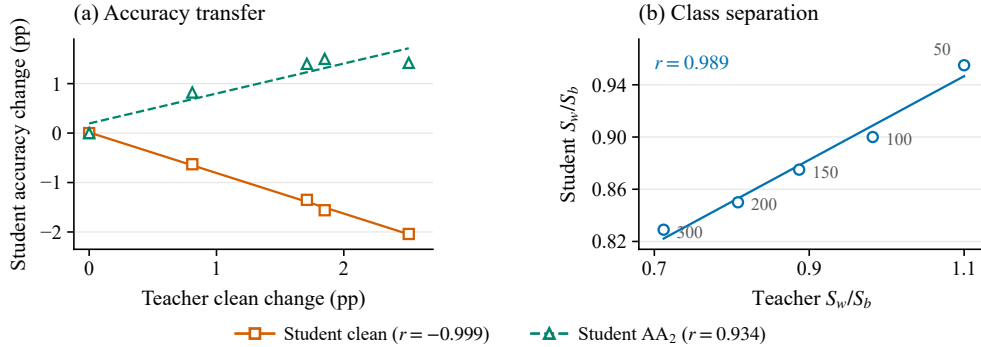


Figure 3: Teacher checkpoint sweep on CIFAR-100 (ResNet-18). (a) Accuracy changes relative to the 50-epoch teacher. (b) Within-to-between scatter ratios S_w/S_b ; lower means stronger class separation, and labels denote teacher epochs. Lines are least-squares fits. Exact values are in Table 10.

The teacher gains 2.51 clean points across the sweep while the student loses 2.04, producing a negative correlation ($r = -0.999$), while student AA_2 rises from 24.38% to 25.80%. Teacher and student scatter ratios decrease together ($r = 0.989$). Thus higher teacher clean accuracy does not guarantee higher student clean accuracy, and changes in teacher class separation are reflected in the student. Five checkpoints from one trajectory establish this association, but do not isolate geometry from the simultaneous changes in initialization and classifier.

We use the 200-epoch checkpoint in subsequent CIFAR-100 experiments. It lies on the late-checkpoint robustness plateau while retaining more student clean accuracy than the 300-epoch teacher. The next section turns the clean target into a training procedure and accounts for the fact that a fixed pixel-space radius perturbs its feature-space loss by different amounts across examples.

4 METHOD

4.1 CLEAN FEATURE ANCHORING

Let Φ_t and h_t be the frozen feature map and classifier of the natural teacher. The student inherits both, $\Phi_s \leftarrow \Phi_t$ and $h_s = h_t$, after which only Φ_s is optimized. For a clean example x , CFA solves

$$\begin{aligned} x_{\text{adv}} &= \arg \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2, \\ \mathcal{L}_{\text{CFA}} &= \mathbb{E}_x [\|\Phi_s(x_{\text{adv}}) - \Phi_t(x)\|_2^2]. \end{aligned} \quad (7)$$

We approximate the inner maximization with randomly initialized PGD against the fixed target $\Phi_t(x)$ and minimize the same discrepancy over the backbone. This adversarial-training stage uses no label loss; inference uses $h_t \circ \Phi_s$. Algorithm 1 gives the complete procedure.

4.2 SENSITIVITY-MATCHED ϵ

A uniform pixel radius need not produce a uniform change in the feature-space objective. We therefore allocate the batch budget according to each example’s clean-point loss sensitivity. For $\ell_{\text{anc}}(x'; x_i) = \|\Phi_s(x') - \Phi_t(x_i)\|_2^2$, define

$$g_i = \|\nabla_{x'} \ell_{\text{anc}}(x'; x_i)\|_1|_{x'=x_i}, \quad \max_{\|\delta\|_\infty \leq \epsilon} \langle \nabla_{x'} \ell_{\text{anc}}, \delta \rangle = \epsilon g_i. \quad (8)$$

The linearized loss change under radius ϵ_i is therefore $\epsilon_i g_i$. For $g_i > 0$, equalizing this quantity while preserving $\sum_i \epsilon_i = N\epsilon$ gives

$$\epsilon_i = N\epsilon \frac{g_i^{-1}}{\sum_j g_j^{-1}}, \quad \text{or more generally} \quad \epsilon_i \propto g_i^{-p}, \quad (9)$$

where $p = 0$ recovers a uniform radius. We use $p = 1$ and an ℓ_2 gradient norm as a sensitivity proxy, measured with one backward pass at clean inputs. The ℓ_1 version implied by Equation (8) gives 0.29 points lower AA and retains the clean-accuracy gain (Appendix E); exact first-order equalization is therefore a motivation for the implemented rule.

We floor the proxy at 10^{-12} , clip its inverse-mean multipliers to $[0.5, 1.5]$, and normalize them to unit mean before setting ϵ_i . The floor gives a uniform allocation when all clean gradients vanish at teacher initialization. Normalization preserves $\sum_i \epsilon_i = N\epsilon$, although the final multipliers can leave the clipping interval. PGD step sizes are scaled by the same multipliers. Appendix E specifies this procedure and an optional exact-box variant. The matched-budget comparison in Table 5 tests whether the sensitivity signal improves on uniform, difficulty-based, and margin-based allocation.

5 EXPERIMENTS

We evaluate the clean–robust trade-off across datasets and architectures, isolate the anchor’s contribution, and distinguish radius-allocation effects from weight averaging and AWP. We then examine feature geometry and attack reliability.

5.1 EXPERIMENTAL SETTINGS

We use CIFAR-10, CIFAR-100 (Krizhevsky et al., 2009), and Tiny-ImageNet-200 (Le & Yang, 2015). The natural teacher is trained on clean data for 200 epochs and then frozen (Table 20); its clean accuracies are 95.32%, 77.66%, and 66.29%, respectively, and its AA is zero on all three datasets. Unless stated otherwise, the ResNet-18 student is initialized from that teacher and trained for 100 epochs with AdamW, a one-cycle schedule, 10-step PGD, and $\epsilon_{\text{train}} = 8.8/255$. The final recipe uses sensitivity matching ($p = 1$), weight averaging, and AWP; exact hyperparameters are in Appendix F.

We evaluate the final-epoch model on the full test set at $\epsilon = 8/255$, reporting clean accuracy, the APGD-CE/APGD-T cascade (AA), and their harmonic mean (NRR). A full four-attack AutoAttack validation on CIFAR-100 yields the same robust accuracy (Appendix H). Baselines use their published training recipes and our evaluation pipeline; the distillation baselines receive our natural teacher. Published results with other teachers and settings are reported separately in Table 26.

5.2 MAIN RESULTS

CFA attains the highest AA and NRR on both CIFAR datasets. On CIFAR-100 it improves over ADR, the closest baseline in NRR, by 1.66 points of AA and 2.00 NRR while also gaining 2.24 clean points. ARREST reaches higher clean accuracy but gives up 6.08 AA points, placing it on a different part of the trade-off. On CIFAR-10, CFA improves over the closest NRR baseline by 1.49 points. The same configuration is evaluated on a larger dataset and network below.

5.3 GENERALIZATION ACROSS DATA AND MODEL SCALE

On Tiny-ImageNet, CFA attains the highest NRR among the measured methods, combining substantially higher clean accuracy than ADR with comparable AA (Table 2). The result is not obtained

Table 1: ResNet-18 results at ℓ_∞ , $\epsilon = 8/255$. All values are measured in our framework. T denotes the teacher assumed by the method: none (–), a natural model (nat.), or the method’s own weight average (self).

Method	T.	CIFAR-10				CIFAR-100			
		Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	–	84.54	43.57	64.06	57.50	57.36	20.34	38.85	30.03
TRADES	–	81.71	49.16	65.44	61.39	55.28	23.55	39.42	33.03
MART	–	80.41	45.56	62.99	58.16	53.86	22.72	38.29	31.96
ARD	nat.	84.58	43.75	64.16	57.67	57.61	20.24	38.93	29.96
RSLAD	nat.	85.15	42.91	64.03	57.06	59.68	21.30	40.49	31.40
AdaAD	nat.	85.39	46.22	65.81	59.98	59.79	23.19	41.49	33.42
AdaAD + IGDM	nat.	85.01	45.50	65.26	59.27	48.25	19.48	33.87	27.75
LBGAT	nat.	–	–	–	–	56.46	26.02	41.24	35.62
ARREST	nat.	85.06	46.42	65.74	60.06	66.97	22.78	44.88	34.00
HAT	nat.	84.59	48.81	66.70	61.90	60.67	24.45	42.56	34.85
ADR	self	83.36	49.72	66.54	62.29	59.93	27.20	43.57	37.42
RPAT++	–	82.41	50.75	66.58	62.82	55.93	27.36	41.64	36.74
CFA (ours)	nat.	84.96	51.74	68.35	64.31	62.17	28.86	45.52	39.42

Table 2: Tiny-ImageNet-200 with ResNet-18. All rows are measured in our framework.

Method	Teacher	Clean	AA	Avg	NRR
PGD-AT	–	49.72	14.45	32.09	22.39
TRADES	–	46.98	15.95	31.46	23.83
MART	–	46.01	16.90	31.46	24.73
ARD	nat.	49.74	14.48	32.11	22.43
RSLAD	nat.	51.26	17.02	34.14	25.56
AdaAD	nat.	51.59	16.79	34.19	25.33
LBGAT	nat.	51.04	18.47	34.75	27.13
ARREST	nat.	59.38	16.06	37.72	25.28
HAT	nat.	52.57	17.28	34.93	26.01
ADR	self	47.66	20.85	34.25	29.01
CFA (ours)	nat.	55.16	20.54	37.85	29.93

by maximizing either endpoint: ARREST remains higher on clean accuracy, whereas ADR is 0.31 points higher on AA. The teacher-checkpoint effect also transfers beyond CIFAR-100: replacing the 200-epoch teacher by its 80-epoch checkpoint changes the student from 55.16/20.54 to 57.08/18.96 clean/AA. Thus the earlier teacher transfers more clean accuracy but less robustness, reproducing the direction of Figure 3 on a third dataset.

Table 3: WideResNet-34-10 at $\epsilon = 8/255$, measured in our framework. The CIFAR-100 CFA entry is a provisional reading of a run whose evaluation is still finishing.

Method	T.	CIFAR-10				CIFAR-100			
		Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	–	86.00	45.71	65.86	59.69	58.97	22.84	40.91	32.93
TRADES	–	85.13	48.99	67.06	62.19	57.35	25.60	41.48	35.40
MART	–	85.29	47.06	66.18	60.65	57.16	22.43	39.80	32.22
ARD	nat.	86.35	45.52	65.94	59.61	59.52	23.05	41.29	33.25
RSLAD	nat.	86.41	45.80	66.10	59.87	60.48	21.63	41.06	31.87
AdaAD	nat.	86.80	46.84	66.82	60.85	61.66	25.62	43.64	36.20
LBGAT	nat.	81.56	51.79	66.68	63.35	59.63	28.55	44.09	38.61
ARREST	nat.	87.54	48.83	68.19	62.69	67.69	24.69	46.19	36.18
HAT	nat.	87.15	53.21	70.18	66.07	62.81	26.91	44.86	37.68
ADR	self	88.64	52.34	70.49	65.82	64.50	29.04	46.77	40.05
CFA (ours)	nat.	88.67	55.29	71.98	68.11	65.00	30.00	47.50	41.05

On CIFAR-10 with WideResNet-34-10, CFA attains the highest AA and NRR among our measured runs (Table 3). Its clean accuracy is close to ADR’s (88.67% versus 88.64%), with 2.95 points higher AA. The CIFAR-100 entry, still provisional, places CFA at 65.00/30.00 clean/AA against ADR’s 64.50/29.04, the closest baseline there. This extends the result to a larger network with the same objective.

5.4 WHAT PRODUCES THE GAIN?

Anchor and label supervision. Replacing the anchor by label cross-entropy under the same base regime reduces clean and AA by 6.69 and 5.91 points (Table 4). Giving cross-entropy its own optimizer and schedule narrows but does not close the gap: CFA remains ahead by 4.63 clean and 4.45 AA points.

Table 4: Objective control on CIFAR-100 with ResNet-18, 100 epochs, no weight averaging or AWP, $p = 0$, and $\epsilon_{\text{train}} = 8.8/255$.

Objective	Clean	AA	NRR
Anchor	61.29	25.36	35.88
Label cross-entropy, same recipe	54.60	19.45	28.68
Label cross-entropy, own recipe	56.66	20.91	30.55

Changing initialization or optimizer and schedule changes NRR by 0.22 and 0.11, respectively (Table 22). With the full stack, label cross-entropy reaches 57.73/26.46 clean/AA versus CFA’s 62.17/28.86; ARREST is the closest of ten recipe-matched baselines in NRR (Table 23). With the same teacher, optimizer, WA, and AWP at 8/255, CFA’s AA advantage over ARREST is 0.70 points at 100 epochs and 1.58 at 150 epochs (Table 18). The latter comparison gives 64.75/27.98 versus 64.66/26.40 clean/AA: higher robustness at nearly equal clean accuracy, supporting CFA’s training design beyond the shared stack.

Regression and probability targets. Logit regression remains effective with WA and AWP. At a uniform training radius of 8.8/255, 100-epoch logit MSE reaches 60.36/28.02 clean/AA against feature regression’s 60.42/28.42 (Table 11). Together with the base-regime comparison in Section 3.2, this shows that effective clean-target regression is not confined to feature space. Teacher-only KL instead reaches 56.97/26.96 at its best tested temperature under the full recipe (Table 25); this separate comparison does not isolate the target from classifier and radius choices.

5.5 RADIUS ALLOCATION AND TRAINING COMPONENTS

Which signal should allocate the radius? Table 5 holds the mean training radius fixed and changes the allocation signal under the full recipe. Sensitivity improves clean/AA by 1.75/0.44 over a uniform radius. To distinguish the signal from the spread of radii, we also preserve the sensitivity rule’s exact set of multipliers but reassign them so easier examples receive larger radii. This reduces AA by 0.80 points. Difficulty and margin alternatives likewise have lower NRR, supporting the use of the anchor’s own input sensitivity. These are signal controls inspired by adaptive-radius methods, rather than reproductions of their complete training algorithms.

Table 5: Radius-allocation signals under the same mean radius, CIFAR-100, ResNet-18, 100 epochs with WA and AWP. $\text{sd}(w)$ measures multiplier dispersion.

Allocation	$\text{sd}(w)$	Clean	AA	NRR
Uniform	0	60.42	28.42	38.66
Sensitivity	0.32	62.17	28.86	39.42
Same multipliers, difficulty order	0.32	60.90	28.06	38.42
Difficulty magnitude	0.39	61.27	28.12	38.55
Logit margin	0.42	61.07	27.79	38.20

Adding weight averaging and AWP. Table 6 follows the additions from the base anchor to the final recipe. Sensitivity matching first raises clean/AA by 1.80/0.38. Weight averaging then adds

1.95 AA points and AWP adds 1.17, with a total NRR gain of 3.54 over the base anchor. These are incremental effects in the stated order; they do not isolate interactions between components. Radius sweeps and WA/AWP comparisons on both CIFAR datasets are in Tables 15 and 16.

Table 6: Component ablation on CIFAR-100 with ResNet-18, 100 epochs, and $\epsilon_{\text{train}} = 8.8/255$. Each row adds one component to the row above.

	Clean	AA	NRR
Anchor ($p = 0$)	61.29	25.36	35.88
+ sensitivity-matched ϵ	63.09	25.74	36.56
+ weight averaging	62.30	27.69	38.34
+ AWP	62.17	28.86	39.42

5.6 WHAT CHANGES IN THE LEARNED GEOMETRY?

Proposition 1 bounds variation within an anchored model, but does not predict a smaller variation than label training. We therefore examine both displacement and class geometry. Under the same true-label PGD protocol, the anchor and label-trained students move by 0.275 and 0.254 of their clean feature norms. These attack-based displacements are not the worst-case diameter O ; they provide no evidence that the anchor’s advantage comes from smaller displacement in this test.

Table 7: Feature geometry under the same true-label PGD attack. Angles are measured on unit-normalized features. *Own* is the sample-to-own-centroid angle; *gap* is the mean angle from each class centroid to its nearest other-class centroid.

	Clean			Adversarial		
	S_w/S_b	Own	Gap	S_w/S_b	Own	Gap
Anchor	0.898	28.1°	28.6°	2.681	33.3°	16.2°
Label CE	4.960	45.4°	21.6°	7.143	47.0°	18.3°

The stronger difference is where those features lie. The anchored model has a $5.5\times$ smaller clean S_w/S_b ratio and a $2.7\times$ smaller ratio under attack; its clean clusters are tighter and their centroids are farther apart (Table 7). Together with the checkpoint sweep, this motivates class organization as a candidate explanation for transfer. The evidence remains observational, and these diagnostics use earlier base-regime models with a trained student head (Appendix C.2); they should not be read as a causal account of the final recipe.

5.7 RELIABILITY AND SCOPE

The full standard AutoAttack suite (APGD-CE, APGD-T, FAB-T, and Square) achieves 28.73% robust accuracy on the CIFAR-100 repeat, identical to the two-attack cascade in this validation. Neither FAB-T nor Square finds additional failures among the 2873 surviving test examples. PGD accuracy also decreases with attack steps and radius, reaching zero at 64/255 (Appendix H). The original repeat evaluation gives 62.00/28.74 clean/AA against 62.17/28.86 for seed 0; two seeds do not establish the significance of small ablation differences (Appendix F). On one RTX 5080, teacher and student training take 0.39 and 2.07 hours, respectively (Table 21).

6 CONCLUSION

Clean Feature Anchoring uses a naturally trained model’s clean representation as the sole target during adversarial training. The fixed anchor jointly controls teacher fidelity and local feature variation, while sensitivity-based radius allocation improves the measured clean–robust trade-off. Experiments across three datasets show gains over label training that persist under changes to the training recipe, with further improvements from weight averaging and AWP. Direct logit regression is also effective, supporting the broader value of regression to a clean teacher target. Finally, higher teacher clean accuracy need not yield higher student clean accuracy, and changes in teacher class separation are

486 reflected in the student. These findings motivate selecting natural teachers for the resulting student
487 trade-off and investigating which aspects of their geometry aid transfer.
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A ALGORITHM

Algorithm 1 Clean Feature Anchoring (CFA)

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1: Input: dataset  $\mathcal{D}$ , radius  $\epsilon$ , attack steps  $K$ , step size  $a$ , exponent  $p$ , clipping thresholds  $[w_{lo}, w_{hi}]$ 
2: Output: robust student  $h_t \circ \Phi_s$ 
3:
4: {Phase 1 — natural self-teacher. The only stage that uses labels.}
5: Train  $f_t = h_t \circ \Phi_t$  on  $\mathcal{D}$  with standard cross-entropy; freeze it
6:
7: {Phase 2 — anchored adversarial training. No labels, no head update.}
8:  $\Phi_s \leftarrow \Phi_t$  {student initialized at the teacher}
9: for epoch = 1 to  $T$  do
10:   for minibatch  $\{x_i\}_{i=1}^N \subset \mathcal{D}$  do
11:      $\phi_i \leftarrow \Phi_t(x_i)$  {teacher read once, at the clean input}
12:      $g_i \leftarrow \max(\|\nabla_x \|\Phi_s(x_i) - \phi_i\|_2, 10^{-12})$  {one backward pass}
13:      $v_i \leftarrow \text{clip}((\bar{g}/g_i)^p, w_{lo}, w_{hi})$ ;  $w_i \leftarrow Nv_i / \sum_j v_j$ 
14:      $\epsilon_i \leftarrow \epsilon w_i$ ;  $a_i \leftarrow aw_i$  { $\sum_i \epsilon_i = N\epsilon$ }
15:      $x_i^0 \leftarrow x_i + \eta_i$ 
16:     for  $k = 0$  to  $K - 1$  do
17:        $x_i^{k+1} \leftarrow \Pi_{B(x_i, \epsilon_i)}(x_i^k + a_i \text{sign } \nabla_x \|\Phi_s(x_i^k) - \phi_i\|_2)$ 
18:     end for
19:      $\mathcal{L} \leftarrow \frac{1}{N} \sum_i \|\Phi_s(x_i^K) - \phi_i\|^2$ 
20:      $\Phi_s \leftarrow \Phi_s - \mu \nabla_{\Phi_s} \mathcal{L}$  { $h_t$  is never updated}
21:   end for
22: end for

```

B TWO PARTIAL RESULTS ON THE ROBUSTNESS SIDE

Natural-teacher guidance is shared with prior work; our question is whether a fixed clean target can support adversarial training without an additional label objective. In CFA, the teacher specifies the target while the student’s inner maximization enforces matching under perturbations. Teacher vulnerability alone therefore does not determine the student’s response: matching a clean target is different from reproducing the teacher at perturbed inputs.

The two results below examine this distinction in restricted settings. The first studies robust and non-robust features in a two-feature model; the second relates anchor error to a classification margin. They supplement the fixed-anchor decomposition, but do not predict the measured accuracy or establish superiority over other regression targets.

B.1 IN THE TWO-FEATURE MODEL, THE NON-ROBUST COEFFICIENT IS ZEROED RATHER THAN SHRUNK

The setting is the two-feature model of Tsipras et al. (2018), which is where the objection is usually posed. One feature x_1 agrees with the label with probability $p = 0.95$ and cannot be attacked. A further d features $z_j \sim \mathcal{N}(\eta y, 1)$ are individually almost uninformative, are jointly close to perfectly informative through their mean $m(z)$, and can all be moved by an ℓ_∞ adversary. A naturally trained network therefore reads $m(z)$: near-perfect clean accuracy, no robustness whatever. That is our teacher’s profile, so $\Phi_t = m(z)$.

Write the student’s feature as $\Phi_s = ax_1 + bm(z)$. Then b is exactly the quantity the objection is about: how much of the teacher’s fragile route the student takes over. If the objection is right, minimizing our objective should leave b large.

Proposition 2. *With the inner maximum exact, $J(a, b) = \mathbb{E}[R^2] + 2\epsilon|b| \mathbb{E}|R| + \epsilon^2 b^2$ with $R = ax_1 + (b - 1)m$, kinked at $b = 0$ with subgradient jump $2\epsilon \mathbb{E}|R(a, 0)| > 0$; above a threshold ϵ_0 the minimizer has $b^* = 0$ exactly.*

Proof. The adversary reaches Φ_s only through m , and only through the mean perturbation $\bar{\delta} = \frac{1}{d} \sum_j \delta_j$, which ranges over $[-\epsilon, \epsilon]$ as $\|\delta\|_\infty \leq \epsilon$. The inner objective is therefore $\max_{|\bar{\delta}| \leq \epsilon} (R + b\bar{\delta})^2 = (|R| + \epsilon|b|)^2$, since a scalar quadratic is maximized at an endpoint and the sign of $\bar{\delta}$ is free. Taking expectations and expanding the square gives the display.

Fix a and differentiate in b . The term $\mathbb{E}[R^2]$ and $\epsilon^2 b^2$ are smooth, while $2\epsilon|b| \mathbb{E}|R|$ contributes $\pm 2\epsilon \mathbb{E}|R|$ as $b \rightarrow 0^\pm$. The one-sided derivatives at $b = 0$ are thus $2\mathbb{E}[R(a, 0) m] \pm 2\epsilon \mathbb{E}|R(a, 0)|$, so $b = 0$ is a minimum along b exactly when

$$|\mathbb{E}[R(a, 0) m]| \leq \epsilon \mathbb{E}|R(a, 0)|. \quad (10)$$

At $b = 0$ the objective in a is $\mathbb{E}(ax_1 - m)^2$, minimized at $a^* = \mathbb{E}[x_1 m] / \mathbb{E}[x_1^2] = q\eta$, and there $\mathbb{E}[R(a^*, 0) m] = q^2 \eta^2 - (\eta^2 + \frac{1}{d}) = -c$ with

$$c = \eta^2 v + \frac{1}{d}, \quad v = 1 - q^2. \quad (11)$$

Equation (10) therefore holds for every

$$\epsilon \geq \epsilon_0 = \frac{c}{\mathbb{E}|q\eta x_1 - m|}, \quad (12)$$

which is the stated threshold. The mechanism is visible in Equation (10): the left side is fixed while the right grows linearly in ϵ , so the kink's slope overtakes the quadratic gain at a finite radius rather than asymptotically. $\square$

Why the coefficient is deleted rather than shrunk. The adversary contributes $2\epsilon|b| \mathbb{E}|R|$ to the loss — linear in $|b|$, not quadratic in b . A quadratic penalty pushes a coefficient toward zero and never arrives; a penalty on $|b|$ is kinked at the origin and sets the coefficient to zero outright once its slope dominates, which is the same distinction that makes the lasso sparse and the ridge not. What produces the kink here is that the student is asked to reproduce a *value* rather than a *route*: the fragile features carry that value on clean data but not inside the ball, so past a certain radius carrying any of them costs more than it pays. The teacher supplies what to match, and the inner maximization discards how the teacher got there.

At $d = 512$ the model's constants give $c = 7.89 \times 10^{-3}$ and $\mathbb{E}|q\eta x_1 - m| = 0.0529$, hence $\epsilon_0 = 0.843 \eta$ — inside the bracket $(0.5\eta, 1.0\eta)$ that the Monte-Carlo minimization returns, and far below the 2η at which the model is usually stated.

Two caveats are worth stating rather than leaving to be found. First, Equation (12) is a condition for $b = 0$ to minimize along b at $a = a^*$; J is not jointly convex, because $|b| \mathbb{E}|R|$ is a product of two convex factors, so global minimality is what the numerical check below supplies and not what the argument proves. Second, the model assumes x_1 is unperturbable and z fully perturbable, so *some* suppression of z is built in before anything is shown. What is not built in is exactness, the threshold, and the ϵ -independence that follows it.

That the exactness is the kink's doing, and not a general consequence of adding an adversary, is visible from dropping it. With the $|b|$ term removed, $J_{\text{quad}} = \mathbb{E}[R^2] + \epsilon^2 b^2$; minimizing out a and writing $u = 1 - b$ collapses it to $u^2 c + \epsilon^2 (1 - u)^2$, so

$$b_{\text{quad}}^* = \frac{c}{c + \epsilon^2}, \quad (13)$$

which is strictly positive at every radius. Against the exact minimizer ($d = 512$, Monte-Carlo, $n = 2 \times 10^5$):

ϵ	b^* (exact)	b_{quad}^* (kink dropped)
0.5η	0.510	0.502
1.0η	0.000	0.202
2.0η	0.000	0.060
4.0η	0.000	0.016

The two columns agree below ϵ_0 and separate above it: the quadratic surrogate shrinks the non-robust coefficient and never deletes it, which is the ordinary picture, while the kinked objective sets it to zero and keeps it there.

Since every ϵ in J multiplies b , at $b^* = 0$ the objective is $\mathbb{E}(ax_1 - m)^2$ and is ϵ -free—past the threshold a larger radius costs the anchor nothing, which matches the saturation measured in Table 15, where AutoAttack and CW at 10/255 are identical to the last decimal to those at 8.8/255 on both datasets.

This model rules out the paper’s effect by construction, so Proposition 2 cannot be the explanation of it: its reliability spectrum has two atoms and no mass near $\eta_j/\epsilon = 1$, where the effect lives. It is a statement about the free half only.

B.2 WHAT ROBUST ACCURACY TRACKS IS THE LOSS MEASURED AGAINST A MARGIN

Writing $\gamma_t(x)$ for the teacher’s margin at x and D_y for the relevant decision-boundary constant, Cauchy–Schwarz gives that $\gamma_t(x) > D_y \cdot L(x)$ implies the student is robustly correct at x . The certificate is vacuous at the scales we train at, but the ratio DL/γ_t tracks both axes across the teacher ladder of Figure 3. What it does not do is predict the size of the effect, and it saturates before AutoAttack does.

C SUPPLEMENTARY ANALYSIS OF THE CLEAN-FEATURE ANCHOR

C.1 TEACHER EVALUATION ON PERTURBED INPUTS

The comparison in Section 3.2 evaluates the teacher at clean inputs. Here we examine why its responses to perturbed inputs may be unsuitable as supervisory targets. Under $\epsilon = 8/255$, the natural teacher’s feature rotates by 63.8° and its norm grows by a factor of 2.45. Its representation can therefore change substantially within the perturbation ball.

As a control, replacing the clean target $\Phi_t(x)$ with $\Phi_t(x')$ makes an exact copy of the teacher, $\Phi_s = \Phi_t$, a zero-loss solution even when the teacher is sensitive to perturbations. This variant yields 0.00% AutoAttack accuracy (Table 8). With a fixed clean target, copying the teacher instead incurs a loss wherever its features change within the ball.

Table 8: Where the teacher is evaluated. CIFAR-100, ResNet-18, 50 epochs; the two rows differ only in the teacher input used by the loss and inner maximization.

Teacher target	Clean	AA	NRR
$\Phi_t(x)$	61.33	26.19	36.71
$\Phi_t(x_{\text{adv}})$	76.11	0.00	0.00

IGDM (Lee et al., 2025) uses the finite difference $f_t(x + \delta) - f_t(x - \delta)$ to approximate the teacher’s response along the perturbation direction. Its accuracy depends on local linearity. The reported Taylor remainder proportion is 0.012 for adversarially trained teachers; we obtain 0.0111 for an adversarially trained ResNet-18 and 0.4763 for the natural teacher (Table 12). For the natural teacher, the finite difference has norm 14.14, compared with $\|f_t(x)\| = 18.36$, and its softmax assigns 0.917 probability to one class. These measurements indicate a less accurate local linear approximation and a highly concentrated target distribution.

We also evaluate published distillation objectives with the natural teacher replacing their robust teacher, using each method’s published recipe (Table 17). All evaluated objectives perform below adversarial training initialized from that teacher. This comparison complements the target sweep in Section 3.2, although differences among training recipes prevent it from isolating the target alone.

C.2 ADDITIONAL EVIDENCE FOR THE ANCHOR OBJECTIVE

The geometry and displacement diagnostics in Section 5.6 use the earlier 100-epoch ladder_p0_100ep anchor and abl_ce_nostack checkpoints. The anchor’s backbone uses raw feature regression, while a detached KD term trains its classifier; this is distinct from the frozen-head models in Table 4. Both diagnostics use true-label PGD with 10 steps at radius 8.8/255. The displacement script samples training batches, whereas the class-geometry script samples test

batches. Consequently, they are complementary descriptive measurements rather than a joint causal test on the final model.

The classification benefit is evaluated in Section 5.4. Without weight averaging, AWP, or the sensitivity-matched radius, the anchor improves clean and AutoAttack accuracy over label cross-entropy by 6.69 and 5.91 points under the same training conditions. The gains are 4.63 and 4.45 points when cross-entropy uses its own optimizer and schedule (Table 4, Table 22). At the full training configuration, the AutoAttack gain over cross-entropy with weight averaging and AWP is 2.40 points. Under a matched attack, the students exhibit similar relative feature displacement, while their class separation differs by a factor of 5.5. This suggests a contribution from the learned geometry, but does not establish class separation as the sole cause of the robustness gain.

The checkpoint ladder in Figure 3 further separates within-class concentration from between-class separation on unit-normalized features. From the 50-epoch to the 300-epoch teacher, the mean angle from a sample to its class centroid decreases from 34.6° to 30.9° , whereas the nearest-other-centroid angle between class means increases from 35.7° to 44.9° . The corresponding students reproduce the latter change—their nearest-centroid separation increases from 25.4° to 29.8° —while their mean within-class angle remains nearly unchanged (27.9° to 28.2°). These measurements motivate interpreting the ladder through between-class geometry, but they remain observational because the teacher’s accuracy and geometry change together. The numerical values underlying the two analysis figures are collected below.

Table 9: Numerical values for the target comparison in Figure 2. $H(q_t^\tau)$ is measured in nats, and NRR is the harmonic mean of clean and AA₂ accuracy.

Target	τ	$H(q_t^\tau)$	Clean	AA ₂	NRR
<i>Teacher-only temperature</i>					
KL	1	0.76	58.26	20.84	30.70
KL	2	2.40	59.33	22.79	32.93
KL	4	4.20	59.39	24.48	34.67
KL	8	4.56	59.47	25.19	35.39
KL	16	4.60	57.78	24.00	33.91
<i>Shared-temperature KD</i>					
KL	4	4.20	60.83	25.05	35.49
KL	8	4.56	60.99	25.25	35.71
KL	16	4.60	61.85	25.39	36.00
KL	32	4.60	61.81	25.50	36.11
Logit regression, fixed head	—	—	62.59	25.69	36.43
Feature anchor, fixed head	—	—	62.72	25.88	36.64

Table 10: Numerical values for the checkpoint trends in Figure 3.

Checkpoint	Teacher		Student		
	Clean	S_w/S_b	Clean	S_w/S_b	AA ₂
50 epochs	75.81	1.100	64.28	0.955	24.38
100 epochs	76.62	0.982	63.65	0.900	25.20
150 epochs	77.52	0.887	62.93	0.875	25.78
200 epochs	77.66	0.808	62.72	0.850	25.88
300 epochs	78.32	0.712	62.24	0.829	25.80

C.3 SCOPE OF THE ANCHOR BOUND

Proposition 1 concerns exact maxima over the perturbation set. A loss evaluated using a finite attack does not certify these maxima. The factor 3 combines $F \leq L$ with $O \leq 2L$; the latter bound is attained when two student features lie on opposite sides of the target, each at distance L .

The argument applies to any fixed target in a normed space, including logits when matched using a norm distance. It explains how one anchor controls two quantities, but does not establish that features are superior to logits. The feature choice is motivated by Section 3.2. Likewise, the bound

does not directly lower-bound robust classification accuracy: this requires information about the classifier and class margins, as discussed in Appendix B.

Joint control also does not imply that fidelity and stability can both be made arbitrarily small over the dataset. When perturbation balls overlap but their teacher targets differ, they impose conflicting constraints on the same student feature. The proposition describes how the objective controls the two errors without eliminating this possible conflict.

D ADDITIONAL RESULTS

The following tables provide additional results supporting the main experiments and supplementary analysis.

Regression under weight averaging and AWP. Table 11 compares logit and feature regression with the same natural teacher, inherited frozen classifier, AdamW optimizer, one-cycle schedule, WA, and AWP. Both use a uniform radius: neither run includes sensitivity-based allocation. Each objective is maximized to generate training perturbations and minimized to update the backbone, so the comparison includes its effect on both stages. Logit regression remains competitive with the training stack; the single-seed difference does not establish a general advantage for feature regression.

Table 11: Fixed-head regression with WA and AWP: CIFAR-100, ResNet-18, 100 epochs, uniform $\epsilon_{\text{train}} = 8.8/255$. Evaluation uses the APGD-CE/APGD-T cascade at $8/255$. Both losses sum squared errors across output coordinates and average over the batch.

Target	Clean	AA
Clean logits	60.36	28.02
Clean features	60.42	28.42

Table 12: Local linearity, which IGDM’s construction requires of the teacher: remainder proportion of the first-order Taylor expansion over an $\epsilon = 8/255$ ball, computed as in IGDM’s Sec. 3.1.

Network	Remainder proportion
IGDM’s robust teachers (LTD / BDM-AT / IKL-AT, published)	0.012 / 0.012 / 0.016
PGD-AT ResNet-18 (ours)	0.0111
Natural teacher (ours)	0.4763
Anchored student, $p = 0$	0.0029
Anchored student, shipped recipe	0.0202

Table 13: What to do with the classifier. CIFAR-100 base regime (50 epochs, raw features, no weight averaging, no AWP, $\epsilon = 8/255$), so that no stack component can absorb the difference.

	Clean	PGD-20	CW	AA	NRR
Logit target only, $\tau = 1$	58.26	22.62	22.79	20.84	30.70
Logit target only, $\tau = 4$	59.39	30.30	26.84	24.48	34.67
Logit target only, $\tau = 16$	57.78	31.57	26.11	24.00	33.91
Feature anchor + head KD, $\tau = 1$	62.34	26.34	26.11	23.93	34.58
Feature anchor + head KD, $\tau = 16$	62.61	32.47	27.32	25.61	36.35
Feature anchor, head untouched	62.72	28.69	27.80	25.88	36.64
head refit, adversarial CE	62.77	29.93	27.66	25.65	36.42
head refit, adv. CE + smoothing 0.1	62.57	30.43	27.55	25.66	36.39
head refit, clean CE	62.70	28.90	27.23	25.15	35.90

Normalization check. Because Section 3.4 analyzes angular geometry, we check whether discarding feature magnitude changes the training conclusion. Normalizing both sides changes NRR by 0.18 with weight averaging and AWP, while raw regression is better in the base regime. We therefore retain raw squared ℓ_2 as the default and use normalized features only for analysis; CFA does not rely on a cosine objective.

Table 14: Effect of feature normalization in the anchor loss. Normalizing both representations to the unit sphere removes their magnitudes and keeps only the angle, for which the squared Euclidean loss satisfies $\|\hat{\Phi}_s - \hat{\Phi}_t\|_2^2 = 2 - 2 \cos \angle(\Phi_s, \Phi_t)$ and is therefore equivalent to cosine loss up to an affine transformation. CIFAR-100, ResNet-18, 100 epochs; within each panel, the runs differ only in how the student and teacher features are normalized. The asymmetric row normalizes the student alone, placing the two sides at different scales, and is included only as a diagnostic.

	Clean	AA	NRR
<i>(a) With the stack: WA, AWP, $\epsilon_{\text{train}} = 8.8/255$</i>			
Raw features (squared ℓ_2)	60.21	28.14	38.35
Both normalized (cosine)	61.41	28.07	38.53
<i>(b) Base regime: no WA, no AWP, $\epsilon_{\text{train}} = 8/255$</i>			
Raw features (squared ℓ_2)	62.40	24.34	35.02
Both normalized (cosine)	61.52	22.90	33.38
Asymmetric: student normalized, teacher raw	61.28	22.98	33.43

Table 15: ϵ_{train} swept with everything else fixed. Evaluation is at $\epsilon = 8/255$ in every cell.

	Clean	PGD-20	CW	AA	NRR
<i>CIFAR-100</i>					
6/255	68.07	28.40	28.16	25.82	37.44
7/255	65.61	30.34	29.64	27.44	38.70
8/255	63.89	31.41	29.83	27.78	38.72
8.8/255 [†]	62.17	32.37	30.93	28.86	39.42
10/255 [†]	59.99	32.82	30.53	28.77	38.89
<i>CIFAR-10</i>					
6/255	89.64	49.29	50.20	47.88	62.42
7/255	88.30	51.43	52.64	50.27	64.07
8/255 [†]	87.22	52.43	53.55	51.15	64.48
8.8/255 [†]	84.96	52.98	53.94	51.74	64.31
10/255 [†]	83.29	53.25	53.88	51.79	63.87

Table 16: Effect of AWP on top of weight averaging across datasets and training radii. Every run uses the sensitivity-matched radius for 100 epochs; the AWP proxy uses $\gamma = 0.005$ after a 10-epoch warmup. Within each row, AWP is the only added component.

	WA			WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
<i>CIFAR-100</i>						
8/255	63.54	26.75	37.65	63.89	27.78	38.72
8.8/255	62.30	27.69	38.34	62.17	28.86	39.42
<i>CIFAR-10</i>						
8/255	86.67	49.20	62.77	87.22	51.15	64.48
8.8/255	85.16	50.15	63.13	84.96	51.74	64.31

Table 17: Published adversarial-distillation objectives supplied with our natural teacher. The first row of each method uses its published recipe; indented rows add our teacher initialization and then our optimizer and schedule, without weight averaging or AWP.

	CIFAR-100			CIFAR-10		
	Clean	AA	NRR	Clean	AA	NRR
ARD	57.61	20.24	29.96	84.58	43.75	57.67
+ teacher initialization	57.96	20.22	29.98	–	–	–
RSLAD	59.68	21.30	31.40	85.15	42.91	57.06
+ teacher initialization	59.87	20.71	30.77	–	–	–
+ our optimizer and schedule	54.93	19.44	28.72	–	–	–
AdaAD	59.79	23.19	33.42	85.39	46.22	59.98
+ teacher initialization	59.75	22.78	32.98	–	–	–
+ our optimizer and schedule	56.51	22.43	32.11	–	–	–
AdaAD + IGDM	48.25	19.48	27.75	85.01	45.50	59.27
+ teacher initialization	49.67	20.15	28.67	–	–	–
PGD-AT, teacher initialization	57.73	26.46	36.29	82.23	50.72	62.74
CFA, no WA/AWP	61.29	25.36	35.88	–	–	–

Table 18: Epoch-matched CFA and full-stack ARREST with the same 200-epoch natural teacher, optimizer, WA, and AWP. The first two panels use a mean training radius of 8/255; the final panel also gives ARREST our radius allocation at 8.8/255. Δ AA is CFA minus ARREST, in percentage points.

Epochs	CFA (ours)			ARREST + WA + AWP			
	Clean	AA	NRR	Clean	AA	NRR	Δ AA
<i>CIFAR-100</i>							
100	63.89	27.78	38.72	64.87	27.08	38.21	+0.70
150	64.75	27.98	39.07	64.66	26.40	37.49	+1.58
<i>CIFAR-10</i>							
50	84.48	50.47	63.19	84.49	50.61	63.30	−0.14
100	87.22	51.15	64.48	85.24	51.52	64.22	−0.37
150	87.37	51.33	64.67	85.79	51.84	64.63	−0.51
200	87.76	51.44	64.86	86.13	51.33	64.33	+0.11
<i>ARREST + our radius, against CFA at the same mean radius</i>							
100, 8.8/255, C100	62.17	28.86	39.42	64.08	27.53	38.51	+1.33
100, 8.8/255, C10	84.96	51.74	64.31	85.93	51.76	64.60	−0.02

Table 19: RPA++ given the rest of our stack. Its own recipe already carries weight averaging (EMA 0.999 from epoch 105 of 200), so the missing component is the AWP proxy, added at $\gamma = 0.005$ with warmup at 10% of the run and evaluated on its own loss; the last row also starts the run at our natural teacher. CIFAR-100, ResNet-18, our reproduction, reported at both the final and the PGD-20-selected weight-averaged checkpoint.

	Final			Best by PGD-20		
	Clean	AA	NRR	Clean	AA	NRR
RPA++ (its own recipe)	57.05	26.61	36.30	55.93	27.37	36.74
+ AWP	57.48	28.05	37.70	57.50	27.92	37.60
+ AWP + teacher initialization	57.83	27.47	37.25	57.80	27.59	37.36

E THE ALLOCATION RULE AND ITS BUDGET SOLVE

E.1 WHERE THE RULE COMES FROM

The objective is optimized over a **pixel** ball, while what it measures is a displacement in feature space, and a fixed pixel radius moves that displacement by very different amounts across samples. The allocation follows from one elementary fact: for a differentiable loss the first-order movement available inside an ℓ_∞ ball of radius e is

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_x \mathcal{L}, \delta \rangle = e \|\nabla_x \mathcal{L}\|_1, \quad (14)$$

the dual norm of ℓ_∞ being ℓ_1 . Equalizing that movement across a batch subject to a fixed total budget $\sum_i \epsilon_i = N\epsilon$ gives $\epsilon_i g_i = \text{const}$, that is $\epsilon_i \propto 1/g_i$ with $g_i = \|\nabla_x \mathcal{L}(x_i)\|_1$. The same allocation is the max-min one: if two samples move by different amounts, shifting budget from the sample that moves more to the sample that moves less raises the minimum, so no unequal assignment is optimal. Taking $\epsilon_i \propto (\bar{g}/g_i)^p$ makes $p = 1$ that allocation, $p = 0$ the uniform-pixel one, and $p \in (0, 1)$ an interpolation.

Two approximations are worth declaring rather than leaving to be found. First, g_i is measured once, by a single backward pass at the *clean* x_i before the attack starts, so it is the sensitivity at the starting point and not along the PGD trajectory. Second, everything above is first order in ϵ . Neither is repaired anywhere in the method; the rule is a budget heuristic with a derivation, not an exactly solved subproblem.

We also do not claim the rule is specific to a directional objective. $\|\nabla_x \mathcal{L}\|_1$ is defined for any loss and the implementation reads whichever loss is configured; on the shipped raw- ℓ_2 anchor it equalizes a feature displacement rather than an angle, which is why the honest name is *sensitivity-matched* ϵ rather than an angular budget. What separates it from IAAT, MMA and CAT is not directionality but the signal: they allocate from a property of the sample, this allocates from the geometry the training loss is written in.

The separation is not one of dispersion, which is worth stating because a rule that simply spread the budget more aggressively would be the simpler explanation of any gain. Measured over training on the shipped configuration, the coefficient of variation of our sensitivity signal has median 0.56, against 0.75 for the difficulty score IAAT and CAT allocate from and 1.33 for the logit margin of MMA. The box of this section compresses all three before they reach the attack, leaving per-sample multipliers of spread 0.32, 0.39 and 0.42 respectively, and the ordering survives the compression. Ours is the least dispersed of the three signals either way, so the comparison in Table 5 is a comparison of which geometry the budget follows.

One discrepancy is ours to declare rather than the reader’s to find. Equation (14) gives the ℓ_1 norm and the implementation reads ℓ_2 , for no better reason than that the cell which fixed the recipe used it and every logged run since had to stay reproducible against that choice. The two are not the same allocation. Measured over training on the shipped configuration, the ℓ_1 signal’s coefficient of variation has median 0.68 against the ℓ_2 signal’s 0.56, and the per-sample multiplier the attack receives has spread 0.34 against 0.32 — so the norms really do distribute the budget differently.

They nonetheless land close together. Training the shipped recipe with `featdir_angeeps_gnorm` set to 1 and nothing else changed gives 62.16/31.97/30.56/28.57 (NRR 39.15) against 62.17/32.37/30.93/28.86 (NRR 39.42): clean accuracy is identical to a hundredth and AutoAttack differs by 0.29, in favour of the norm we ship.

We do not call that a tie. It is twice the seed-to-seed spread of Appendix F, and it is the same order of magnitude as what the allocation buys in the first place: against the uniform radius, $p = 1$ gains +1.75 clean and +0.44 AutoAttack at ℓ_2 and +1.74 and +0.15 at ℓ_1 . So the rule’s benefit survives the norm in direction and almost entirely on the clean axis, while most of its AutoAttack half does not. The honest statement is that the norm is a real choice which we made before we could compare, that the comparison favours what we ship, and that a reader preferring the derived ℓ_1 would still get the clean gain and a third of the robust one.

The permuted row of Table 5 deserves its construction spelled out, since it is what makes the comparison a comparison of signals rather than of budgets. It takes the multiset of weights the gradient

rule produces — the same values, the same clip $[0.5, 1.5]$, the same restored mean — and only permutes which example receives which, so that the larger radii go to the easier examples, which is the direction IAAT and CAT assign them. Reordering costs 0.80 AutoAttack against the rule it reorders and lands below even the uniform row on AutoAttack, CW and NRR alike.

E.2 CLIPPING AND RESTORING THE MEAN RADIUS

The reported configuration floors the gradient norm at 10^{-12} and computes $v_i = \text{clip}((\bar{g}/g_i)^p, 0.5, 1.5)$, followed by $w_i = Nv_i / \sum_j v_j$ and $\epsilon_i = \epsilon w_i$. PGD uses step size $a_i = aw_i$. This preserves the batch mean radius exactly. The clipping thresholds apply before normalization, so they are not hard bounds on the final multipliers. When all gradients vanish, the floor makes all g_i equal and the allocation uniform.

Optional exact-box variant. The implementation also supports an alternative that constrains the final multipliers. It is not enabled in the reported default configuration.

With $r_i = (\bar{g}/g_i)^p$ and box $[w_{\text{lo}}, w_{\text{hi}}] = [0.5, 1.5]$, the radii that satisfy both the box and the budget exactly are $\epsilon_i = \epsilon \cdot \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}})$ with t the solution of

$$\sum_{i=1}^N \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}}) = N. \quad (15)$$

The left side is continuous and nondecreasing in t from Nw_{lo} to Nw_{hi} , so a root exists and bisection finds it; when no clip is active it reduces to $t = N / \sum_i r_i$, that is, to restoring the mean. Clipping first and rescaling afterwards, the obvious alternative, does not preserve the box—the rescaling moves the clipped entries back out of it.

F EXPERIMENTAL DETAILS

Table 20: Natural teacher checkpoints used by the shipped models. The teachers are trained only on clean examples and have zero adversarial accuracy.

Dataset	Architecture	Epochs	Clean
CIFAR-10	ResNet-18	200	95.32
CIFAR-100	ResNet-18	200	77.66
Tiny-ImageNet	ResNet-18	200	66.29

Every number we report is the **final** epoch, never a checkpoint selected on a robustness metric. This matters when reading Table 26: RPAT selects on PGD-20 and ADR on PGD-10, so those rows are upper envelopes of a training run where ours is its endpoint.

Teacher. One naturally trained network per dataset, no adversarial examples anywhere in its training, and the student is warm-started from the same checkpoint. ResNet-18, SGD at learning rate 0.1 under a one-cycle schedule, batch size 128, no augmentation beyond the standard crop and flip, for 200 epochs on each dataset. The resulting teachers are summarized in Table 20. The teacher’s own robustness is 0 — it is never attacked and is not meant to survive attack; Figure 3 is the ablation over how long it is trained.

Student. The three shipped cells differ only where the dataset forces them to.

	CIFAR-10	CIFAR-100	Tiny-ImageNet-200
Backbone	ResNet-18	ResNet-18	ResNet-18
Optimizer	AdamW	AdamW	AdamW
Learning rate / schedule	0.021, one-cycle	0.021, one-cycle	0.021, one-cycle
Epochs	100	100	100
Batch size	128	128	128
Training radius ϵ_{train}	8.8/255	8.8/255	8.8/255
Inner attack	PGD-10, step 2/255	PGD-10, step 2/255	PGD-10, step 2/255
Allocation exponent p	1	1	1
Radius box $[w_{\text{lo}}, w_{\text{hi}}]$	[0.5, 1.5]	[0.5, 1.5]	[0.5, 1.5]
Classifier head	frozen	frozen	frozen
Weight averaging (start, decay)	0.2, 0.999	0.2, 0.999	0.2, 0.999
AWP (γ , warmup)	0.005, 10 ep	0.005, 10 ep	0.005, 10 ep
Teacher checkpoint	clean, 200 ep	clean, 200 ep	clean, 200 ep

The head being frozen is what removes τ and β from the method: with the classifier inherited and never updated, the head-KD term of earlier drafts has no parameters to reach and is switched off entirely. p is not tuned. $\epsilon_{\text{train}} = 8.8/255$ is the one value swept, in Table 15, and evaluation is at 8/255 in every cell regardless.

Evaluation. We report the APGD-CE/APGD-T cascade at ℓ_∞ radius $\epsilon = 8/255$ on the full test set. A dedicated validation of the CIFAR-100 seed-1 model uses all four attacks in standard AutoAttack; FAB-T and Square leave the two-attack result unchanged (Appendix H). PGD-20 and CW_∞ are also reported (Appendix J), and NRR is the harmonic mean of clean and robust accuracy. Where a table reports a method we ported rather than quoted, the port runs on its own published recipe — HAT and LBGAT both collapse under our schedule and are given theirs — and only the evaluation is ours.

Repeated training. Each reported row is an individual run. For the final CIFAR-100 configuration, seed 1 gives 62.00/32.49/30.82/28.74 clean/PGD-20/CW/AA, compared with 62.17/32.37/30.93/28.86 for seed 0. These two runs provide a limited reproducibility check, not a variance estimate or a significance threshold for other comparisons. In particular, small differences between regression targets or nearby operating points require further repetitions.

Compute, teacher included. A single RTX 5080. One student cell is about 2.2 hours on CIFAR-100 at 100 epochs, of which the AutoAttack evaluation is roughly 7 minutes; Tiny-ImageNet is longer in proportion to its test set.

The teacher has to be counted, since calling it free would otherwise hide a training run. It is not free, but it is cheap: a clean epoch has no inner maximization, so it costs 6.8 seconds against 74.5 for an adversarial one, and 200 clean epochs come to 0.39 hours. The teacher is therefore 19% of one student run and is amortized over every cell that reads it, the teacher ladder of Figure 3 included. Table 21 puts the total against the baselines measured on the same machine.

Table 21: Wall-clock training cost on one RTX 5080, CIFAR-100, ResNet-18, as logged by the runs reported in this paper. Evaluation is excluded. Adversarial distillation additionally requires an adversarially trained teacher, which is not counted in its row and is the more expensive artifact.

Method	Epochs	Teacher (h)	Student (h)	Total (h)
Natural teacher (ours, reused)	200 clean	–	0.39	0.39
CFA (ours)	100 adv.	0.39	2.07	2.46
Label CE at our regime	100 adv.	0.39	3.22	3.61
PGD-AT, initialized at the teacher	100 adv.	0.39	1.72	2.11
ARD	100 adv.	0.39	1.52	1.91
RSLAD	100 adv.	0.39	1.71	2.10
AdaAD	100 adv.	0.39	2.79	3.18
LBGAT	100 adv.	joint	1.88	1.88
HAT	50 adv.	0.39	1.10	1.49
ADR	200 adv.	self	3.52	3.52

Two readings follow. The anchor is not bought with compute, since it costs less end to end than label cross-entropy at the same regime and about what PGD-AT costs once the teacher is included. And the comparison with adversarial distillation is favourable in a way the table understates, because those rows assume a robust teacher that the current literature obtains from a WideResNet trained on generated data, which is more expensive than every row here combined.

G THE RECIPE CONTROLS

Unless explicitly ablated, CFA warm-starts from the natural teacher and trains with AdamW at 0.021 under a one-cycle schedule, while the baselines of Table 1 run at their authors’ recipes. Two objections follow from that asymmetry, and Table 22 answers both. The first is that our recipe is what wins; the second is that the baselines would win with it. Every row is measured here at the base regime, which carries no weight averaging, no AWP and a uniform radius, on CIFAR-100 with ResNet-18 for 100 epochs.

Table 22: The recipe controls. *Ours* is AdamW at 0.021 under a one-cycle schedule with the teacher warm start; *theirs* is SGD at 0.1 decayed by ten at epochs 70 and 90, which is PGD-AT’s schedule here and the one the distillation baselines use. The training radius and every other field are held fixed within each pair.

Objective	Recipe	Clean	AA	Avg	NRR
Anchor	ours, teacher init	61.29	25.36	43.33	35.88
Anchor	ours, random init	60.81	25.21	43.01	35.66
Anchor	theirs, teacher init	59.65	25.54	42.60	35.77
Label cross-entropy	ours, teacher init	54.60	19.45	37.03	28.68
Label cross-entropy	theirs, teacher init	56.66	20.91	38.79	30.55
AdaAD	ours, teacher init	56.51	22.43	39.47	32.11
AdaAD	theirs, random init	59.79	23.19	41.49	33.42
RSLAD	ours, teacher init	54.93	19.44	37.19	28.72
RSLAD	theirs, random init	59.68	21.30	40.49	31.40

Table 23: Every baseline given our optimizer, schedule, natural warm start, training radius, weight averaging, and AWP proxy while retaining its own objective. CIFAR-100, ResNet-18.

Method	Own recipe			+ ours			Δ NRR
	Clean	AA	NRR	Clean	AA	NRR	
PGD-AT	57.36	20.34	30.03	57.73	26.46	36.29	+6.26
TRADES	55.28	23.55	33.03	55.33	25.26	34.69	+1.66
MART	53.86	22.72	31.96	50.48	25.74	34.09	+2.14
ARD	57.61	20.24	29.96	58.04	26.44	36.33	+6.37
RSLAD	59.68	21.30	31.40	59.82	25.04	35.30	+3.91
AdaAD	59.79	23.19	33.42	58.37	26.71	36.65	+3.23
AdaAD + IGDM	48.25	19.48	27.75	52.89	25.04	33.99	+6.23
ARREST	66.44	22.99	34.16	63.65	27.48	38.39	+4.23
HAT	60.67	24.45	34.85	55.42	19.56	28.91	−5.94
LBGAT	56.46	26.02	35.62	<i>collapses to chance</i>			−
CFA (ours)	−			62.17	28.86	39.42	

Changing initialization or schedule changes the anchor’s NRR by 0.22 and 0.11, respectively. Cross-entropy improves by 1.87 NRR under its own schedule, so Table 4 includes that stronger baseline. The anchor remains ahead under both tested schedules. These comparisons support an objective-level contribution within the tested recipes, without establishing significance from single runs or claiming exhaustive tuning of the alternatives.

H GRADIENT MASKING

We validate the reported robustness with the full standard AutoAttack suite and additional gradient-masking diagnostics (Athalye et al., 2018). The evaluated model is the CIFAR-100 seed-1 repeat of

the final configuration, with 62.00% clean accuracy and 28.74% in its original two-attack evaluation. In the dedicated validation on all 10000 test images, both the two-attack cascade and the full four-attack suite yield 28.73% (Table 24). This establishes that FAB-T and Square do not reduce robust accuracy further for this checkpoint; it does not substitute for a four-attack evaluation of other checkpoints.

Table 24: Gradient-masking checks, all 10000 test images, $\epsilon = 8/255$. The first block runs each attack *independently* from the clean test set, which is what compares their strengths; the second runs them as AutoAttack does, each attack seeing only the points the previous ones left standing. The teacher-transfer row builds PGD-20 examples on the natural teacher and evaluates them on the student, which shares the teacher’s classifier.

Check	Setting	Accuracy
<i>white box against black box</i>		
APGD-CE	white box, untargeted	31.80
APGD-T	white box, targeted	28.83
FAB-T	white box, boundary	29.16
Square	black box , 5000 queries	34.82
<i>as a cascade</i>		
APGD-CE $\rightarrow$ APGD-T	two-attack validation	28.73
$\rightarrow$ FAB-T $\rightarrow$ Square	AutoAttack’s standard version	28.73
<i>radius sweep, PGD-20</i>		
$\epsilon = 2/255$		54.62
$\epsilon = 4/255$		47.03
$\epsilon = 8/255$		32.46
$\epsilon = 12/255$		20.88
$\epsilon = 16/255$		11.94
$\epsilon = 32/255$		1.05
$\epsilon = 64/255$		0.00
<i>step sweep, $\epsilon = 8/255$</i>		
PGD-10 / 20 / 50 / 100		33.17 / 32.57 / 32.30 / 32.24
<i>transfer</i>		
Natural teacher $\rightarrow$ student	PGD-20 built on the teacher	60.42

Square leaves 5.99 points more accuracy than the strongest individual white-box attack, APGD-T. In the standard cascade, neither FAB-T nor Square successfully perturbs any of the 2873 examples that survive APGD-CE and APGD-T, leaving robust accuracy unchanged at 28.73%. PGD accuracy decreases monotonically over the tested radii, reaching zero at 64/255, and decreases with the number of steps, with smaller changes beyond 50. Together, these checks provide no indication of gradient masking under the evaluated attacks, without establishing worst-case robustness.

The transfer row is worth reading separately. The student inherits the teacher’s classifier and is initialized at the teacher’s features, so adversarial examples built on the teacher might be expected to carry over; they cost the student 1.58 points against its clean accuracy. Sharing a head is therefore not sharing a vulnerability, which is the same conclusion Section 5.4 reaches from class separation: what the anchored student changes is the position of its features, and the teacher’s own adversarial directions do not point at the student’s boundaries.

I NOTES ON THE BASELINES

Several rows of Table 1 run a specific variant of the method they name, or are grouped in a way that needs saying. This section is those decisions; the table carries none of them.

LBGAT, and why its CIFAR-10 cell is empty. The authors release a training script for CIFAR-100 only, and their CIFAR-10 entries are released as pretrained WideResNet checkpoints. Running their CIFAR-100 recipe on CIFAR-10 diverges here within thirty steps: the squared error is reduced with mean over batch and classes, so a ten-class problem receives ten times the gradient of a hundred-class one at equal logit error, and $\text{lr} = 0.1$ is above the stability threshold that follows. Lowering the learning rate trains (at 0.01 the natural branch reaches 49% within 180 steps), but a learning rate

we chose is not their recipe, so the cell is left empty rather than filled with a number of our own construction. LBGAT trains its natural branch jointly with the robust one, so unlike the distillation rows its teacher is not obtained for free, and we group it accordingly. The two datasets run different variants because the authors publish different ones: CIFAR-100 is LBGAT6 and CIFAR-10 is LBGAT0, their release lists CIFAR-10 checkpoints under LBGAT0 alone, and $\beta = 6$ is their CIFAR-100 setting. We report the configuration each dataset has a published counterpart for rather than forcing one β across both.

HAT uses a naturally trained network only to *label* helper points beyond the ball, never as a target, so it belongs with the trade-off methods and not the distillation ones. It is run on its own one-cycle schedule rather than ours, which collapses it to chance on both datasets — the same allowance we make for LBGAT. On CIFAR-10 the port reproduces HAT’s published no-extra-data figure to within 0.31 clean and 0.27 AutoAttack (84.90/49.08 reported, ResNet-18), which is what licenses reporting its CIFAR-100 row at all: HAT publishes no CIFAR-100 result without additional data, its 61.50/28.88 being trained with DDPM-generated samples, so that column has no published figure to be read against.

ADR is run on separate hardware and its row is filled from there. Two things have to hold when it is. ADR reports two configurations — AT+ADR at 56.10/26.87 on CIFAR-100 and AT+WA+AWP+ADR at 57.36/28.50 — and the row has to name which, the second being the one component-matched to our recipe, which carries both weight averaging and AWP. ADR is also self-distillation: its teacher is an exponential moving average of the student, no external network is read, and the student trains from scratch, so its row inherits nothing of our natural teacher.

What the published tables are blocked by. Table 26 and Table 27 separate their rows by network, and the separations are not cosmetic. RPAT’s “ResNet-18” is ResNet(Bottleneck, [2, 2, 2, 2]) in its released code, 14.35M parameters against the 11.27M of the standard BasicBlock network used by us and by ADR; they also ship a `pre_resnet18` at 11.27M and use PreActResNet-18 explicitly elsewhere, so the distinction is theirs. HAT’s Tiny-ImageNet row is PreActResNet-18: ADR lists it under ResNet-18, but the figures are quoted from HAT’s own Table 9, whose caption says PreAct, and HAT’s code refuses anything else on that dataset — same parameter count as ours, different network, and its two companions from that table are shown beside it. F²AT’s plain SAT baseline reaches 31.52/9.37 where ADR’s plain AT reaches 45.87/18.06, a gap no difference of method explains, so its rows are separated too. TE’s architecture we could not verify — it is ADR’s attribution to a paper we do not have — and it is marked accordingly. ADR trains Tiny-ImageNet for 80 epochs against our 100, which favours us and is stated here rather than left implicit. LBGAT reports that dataset without AutoAttack and B-MTARD reports it on MobileNet-v2, so neither appears; CURE, ARREST and Generalist++ do not report it at all.

IGDM is reported at $\alpha = 1$. Its published $\alpha = 20$ diverges to chance accuracy on both datasets when the teacher is natural, as documented in Table 17.

Teacher-only KL with the training stack. This diagnostic combines probability-target distillation with WA and AWP; it is not a reproduction of a published method. Temperature is applied only to the teacher, and the best tested stacked setting, $\tau = 4$, gives 56.97/26.96 clean/AA; $\tau = 16$ gives 42.45/20.10. The comparison concerns this KL training design, not all logit-based supervision. In particular, fixed-head logit MSE is evaluated separately in Table 11, where it remains competitive with feature regression.

J PER-ATTACK METRICS

Table 25 gives every attack we measure for our own cells. The main tables report clean accuracy and AutoAttack because those are the two axes the trade-off is defined on, and because PGD-20 is not a faithful reading of this method.

Why PGD-20 is not the metric here. The student inherits the teacher’s classifier and never retrains it. PGD-20 ascends the cross-entropy *through that classifier*, so it reads a head that was fitted for

Table 25: Per-attack accuracy (%) for our cells. ResNet-18, $\epsilon = 8/255$, final-epoch weight-averaged model, full test set.

Dataset	Cell	Clean	FGSM	PGD-20	CW	AA	NRR
CIFAR-100	CFA, head frozen (shipped)	62.17	36.91	32.37	30.93	28.86	39.42
	the same cell at seed 1	62.00	36.79	32.49	30.82	28.74	39.27
	ℓ_1 gradient norm	62.16	36.48	31.97	30.56	28.57	39.15
	CFA, head-KD term retained	62.35	38.80	36.26	30.65	28.68	39.29
	CFA, $\epsilon_{tr} = 6/255$	68.07	35.88	28.40	28.16	25.82	37.44
	CFA, $\epsilon_{tr} = 7/255$	65.61	36.32	30.34	29.64	27.44	38.70
	CFA, $\epsilon_{tr} = 8/255$	63.89	36.67	31.41	29.83	27.78	38.72
	CFA, $\epsilon_{tr} = 10/255$	59.99	36.50	32.82	30.53	28.77	38.89
	CFA, no WA / AWP	61.29	33.97	28.23	27.19	25.36	35.88
	Logit anchor, our stack	56.97	35.82	33.68	28.94	26.96	36.60
CIFAR-10	RPAT++ (our reproduction)	55.93	34.46	32.44	29.37	27.36	36.74
	CFA, $\epsilon_{tr} = 8.8/255$ (shipped)	84.96	59.58	52.98	53.94	51.74	64.31
	CFA, $\epsilon_{tr} = 6/255$	89.64	59.37	49.29	50.20	47.88	62.42
	CFA, $\epsilon_{tr} = 7/255$	88.30	59.59	51.43	52.64	50.27	64.07
	CFA, $\epsilon_{tr} = 8/255$	87.22	59.92	52.43	53.55	51.15	64.48
	CFA, $\epsilon_{tr} = 10/255$	83.29	58.98	53.25	53.88	51.79	63.87
	RPAT++ (our reproduction)	82.41	59.73	55.72	52.88	50.75	62.82

a different feature distribution; AutoAttack, which includes a targeted APGD, and CW_∞ , which attacks the logit margin directly, do not depend on that fit in the same way. The consequence is measurable on a controlled pair. The first two rows of Table 25 share a backbone objective, a schedule and an attack, and differ only in whether the classifier is refit:

	PGD-20	CW	AA
head-KD term retained	36.26	30.65	28.68
head frozen	32.63	30.66	28.77
difference	-3.63	+0.01	+0.09

Changing the classifier moves PGD-20 by 3.63 points and moves CW and AutoAttack by hundredths. The same pattern recurs across our ablations—the head-temperature sweep, the label-smoothing sweep, the directional-versus-raw comparison, and the radius sweep all show PGD-20 moving against CW and AutoAttack. We therefore report PGD-20 for completeness and read the trade-off on clean accuracy and AutoAttack, which is also what ADR and ARREST report in their main tables.

K PUBLISHED RESULTS, FOR CONTEXT

Table 1 contains only numbers we measured ourselves. Table 26 collects the published figures for the same and neighbouring methods, so that our runs can be placed against the literature without the two being mixed in one column. These are quoted as reported and were produced by other codebases, other schedules and other checkpoint-selection rules—RPAT selects its best checkpoint by PGD-20 and ADR by PGD-10, where every number of ours is the final epoch—so they should be compared within a source rather than across.

Read against Table 1 and restricted to the ResNet-18 blocks, which are the ones our numbers are comparable with, our CFA row exceeds every published Avg and NRR on both datasets: 62.17/28.86 at Avg 45.52 on CIFAR-100 and 84.96/51.74 at Avg 68.35 on CIFAR-10.

The restriction is not a convenience. The WideResNet-34-10 block exceeds ours on both criteria—RPAT++ reaches Avg 70.87 and NRR 67.30 there—because a 48.3M network is a different model, not a better method, and Table 3 is where that comparison belongs. The same caution applies to the PreActResNet-18 blocks in the other direction: RPAT++ falls to 82.63/51.00 there, and reading that as a defeat for RPAT would be the identical error. We state all of this as a comparison across codebases and architectures rather than as a controlled result, which is what Table 1 is for.

Table 26: Published results, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. **The architecture changes between blocks** and is named in each heading; only the ResNet-18 blocks are comparable with Table 1. These were produced by other codebases, schedules and checkpoint-selection rules, so they should be compared within a source rather than across.

Method	Teacher	Clean	AA	Avg	NRR	Source
<i>CIFAR-100, ResNet-18</i>						
PGD-AT	–	56.56	25.02	40.79	34.69	RPAT
TRADES	–	55.39	24.51	39.95	33.98	RPAT
MART	–	49.83	25.00	37.41	33.30	RPAT
Consistency-AT	–	58.53	25.39	41.96	35.42	RPAT
Consistency-AT + RPAT	–	60.33	26.31	43.32	36.64	RPAT
F ² AT	–	54.19	23.24	38.71	32.53	F ² AT
Generalist++	co-tr.	62.97	23.96	43.47	34.71	Generalist++
ADR (AT + ADR)	self	56.10	26.87	41.48	36.34	ADR
ADR (AT + WA + AWP + ADR)	self	57.36	28.50	42.93	38.08	ADR
<i>CIFAR-10, ResNet-18</i>						
PGD-AT	–	82.78	44.63	63.71	57.99	CURE
TRADES	–	82.41	48.37	65.39	60.96	CURE
MART	–	80.70	47.49	64.10	59.79	CURE
ST-AT	–	83.10	50.50	66.80	62.82	CURE
IAD	robust	80.63	50.17	65.40	61.85	CURE
Consistency-AT + RPAT	–	84.12	48.98	66.55	61.91	RPAT
Generalist++	co-tr.	89.09	46.07	67.58	60.73	Generalist++
ARREST	natural	86.63	46.14	66.39	60.21	ARREST
CURE	self	86.76	49.69	68.23	63.19	CURE
ADR (AT + ADR)	self	82.41	50.38	66.40	62.53	ADR
<i>CIFAR-10, PreActResNet-18</i>						
WA (Izmailov et al., 2018)	–	83.50	49.89	66.70	62.46	RPAT
AWP (Wu et al., 2020)	–	81.11	50.09	65.60	61.93	RPAT
KD + SWA (Chen et al., 2021)	robust	84.06	49.82	66.94	62.56	RPAT
TE (Dong et al., 2022)	–	82.04	50.12	66.08	62.23	RPAT
ReBAT (Wang et al., 2023)	–	82.09	50.72	66.41	62.70	RPAT
RPAT++ (Liu et al., 2025)	–	82.63	51.00	66.82	63.07	RPAT
<i>CIFAR-100, PreActResNet-18</i>						
WA (Izmailov et al., 2018)	–	57.26	25.83	41.55	35.60	RPAT
AWP (Wu et al., 2020)	–	54.10	25.16	39.63	34.35	RPAT
KD + SWA (Chen et al., 2021)	robust	57.17	25.66	41.42	35.42	RPAT
TE (Dong et al., 2022)	–	56.41	25.84	41.13	35.44	RPAT
ReBAT (Wang et al., 2023)	–	56.13	27.60	41.87	37.00	RPAT
RPAT++ (Liu et al., 2025)	–	56.84	27.68	42.26	37.23	RPAT
<i>CIFAR-10, WideResNet-34-10</i>						
WA (Izmailov et al., 2018)	–	87.66	52.65	70.16	65.79	RPAT
MMA (Ding et al., 2020)	–	87.80	43.10	65.45	57.82	RPAT
AWP (Wu et al., 2020)	–	85.63	53.32	69.48	65.72	RPAT
KD + SWA (Chen et al., 2021)	robust	87.45	53.59	70.52	66.46	RPAT
TE (Dong et al., 2022)	–	85.97	52.88	69.43	65.48	RPAT
ReBAT (Wang et al., 2023)	–	85.25	54.78	70.02	66.70	RPAT
ADR (Wu et al., 2024)	self	84.67	53.25	68.96	65.38	RPAT
CURE (Gowda et al., 2024)	self	87.05	52.10	69.58	65.19	RPAT
RPAT++ (Liu et al., 2025)	–	86.76	54.97	70.87	67.30	RPAT

Table 27: Published Tiny-ImageNet-200 results, ResNet-18, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. Our row is above all of them on both axes. The blocks are separated because the networks are not the same, and Appendix I says how each differs.

Method	Clean	AA	Avg	NRR	Source
CFA (ours)	55.16	20.54	37.85	29.93	—
AT + WA + ADR	48.55	20.23	34.39	28.56	ADR
AT + WA + AWP + ADR	48.27	20.12	34.20	28.40	ADR
TRADES + WA + AWP + ADR	51.38	19.48	35.43	28.25	ADR
TRADES + WA + ADR	51.99	19.17	35.58	28.01	ADR
AT + WA + AWP	48.61	19.58	34.09	27.92	ADR
AT + WA	49.10	19.30	34.20	27.71	ADR
TE*	47.46	18.29	32.88	26.40	ADR
AT	45.87	18.06	31.96	25.92	ADR
TRADES	48.49	17.35	32.92	25.56	ADR
<i>PreActResNet-18 — same parameter count, different network</i>					
HAT	52.60	18.14	35.37	26.98	HAT
TRADES	48.25	17.17	32.71	25.35	HAT
AT	47.76	17.92	32.84	26.02	HAT
<i>Bottleneck ResNet-18 (14.35M) and an unstated setup — see caption</i>					
Consistency-AT + RPAT	49.74	18.84	34.29	27.33	RPAT
PGD-AT + RPAT	47.68	17.77	32.73	25.89	RPAT
Consistency-AT	46.54	17.60	32.07	25.54	RPAT
MART	39.70	17.18	28.44	23.98	RPAT
F ² AT	40.54	13.13	26.84	19.84	F ² AT
SEAT	35.51	11.37	23.44	17.22	F ² AT
SAT	31.52	9.37	20.45	14.45	F ² AT

L THE CURE REPRODUCTION

Seven runs of the official CURE repository, from its own code and our shared CIFAR data, none of which reproduce the published 86.76/49.69 on CIFAR-10. They fall short on *different* axes: the closest on standard accuracy reaches 86.11 at AA 40.65, the closest on robustness AA 45.63 at clean 81.15, while the published pair has both. Running the repository unmodified gives 65.03/29.74. Computing its RGP prominence mask on absolute values raises AA by 15.9 points (29.74 $\rightarrow$ 45.63), which localizes the discrepancy to that step without closing it.

We re-evaluated all five surviving checkpoints with our own pipeline and it agrees with the repository’s own evaluation to within 0.05 AutoAttack on every one, so the gap is in training rather than in measurement. For contrast, RPAT++ reproduces to within 0.31 AutoAttack of its published numbers on both datasets under the same conditions. The CURE row of Table 26 is therefore the paper’s own number, and CURE is omitted from Table 1.

M A CORRECTION TO THE AWP PROXY, AND WHAT IT MOVED

AWP ascends an auxiliary objective in weight space before each descent step. In our implementation the closure used for that ascent had drifted from the closure used for descent, in two ways. It retained a head-distillation term that `freeze_head` removes from training, so AWP was maximizing a quantity partly composed of a term the model never optimizes; and a detach intended for the normalized feature was applied on a path that feeds the unnormalized one. Measured on CIFAR-100 before the fix, the spurious term was 7.15 against the anchor’s 4.15 and contributed 40.8% of the AWP backbone gradient, at cosine 0.918 to the correct direction. Separately, the head was frozen *after* the proxy was copied, which is harmless only because the proxy’s first step is delayed by the AWP warmup.

We report this because it is the kind of defect that changes a paper’s numbers, and because here it did not. Re-running the shipped recipe with both closures agreeing gives 62.17/28.86 against 62.65/28.77 on CIFAR-100 and 84.96/51.74 against 85.58/51.79 on CIFAR-10: NRR moves by 0.01 and 0.22, no ordering in any table changes, and clean accuracy moves more than robustness

does. AWP is evidently insensitive to a large, well-aligned perturbation of its ascent direction, which is consistent with its role here as a regularizer of sharpness rather than a precisely targeted objective. All numbers reported for the shipped recipe are post-correction. Two exceptions are stated where they appear: the head-freeze comparison in Appendix J and the ϵ_{train} sweep in Table 15 were measured before it. In each, every arm shares the defect, so the comparison is internally consistent even though its absolute values differ from the corrected ones by a few hundredths.

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